Gamma and logGamma prior

Parametrization

The Gamma distribution has density

$$\pi(\tau) = \frac{b^a}{\Gamma(a)} \tau^{a-1} \exp(-b\ \tau) \tag{1}$$

for $\tau > 0$ where:

$a > 0$ is the shape parameter, and

$b > 0$ is the inverse-scale parameter.

The mean of τ is a/b and the variance is a/b^2, and we denote this distribution Gamma(a, b). The variable θ has a *logGamma*(a, b) distribution, if $\theta = \log(\tau)$ and τ is Gamma(a, b) distributed.

Specification

The Gamma and logGamma prior for the hyperparameters is specified as

```
f(<whatever>, hyper = list(<theta> = list(prior="gamma", param=c(<a>,<b>))))
```

or

```
f(<whatever>, hyper = list(<theta> = list(prior="loggamma", param=c(<a>,<b>))))
```

In the case where there is one hyperparameter for that particular f-model. In the case where we want to specify the prior for the hyperparameter of an observation model, for example the Gaussian, the the prior spesification will appear inside the control.family()-argument; see the following example for illustration.

Example

In the following example we estimate the parameters in a simulated example with gaussian responses and assign the hyperparameter θ (the precision parameter), a logGamma prior with parameters $a = 0.1$ and $b = 0.1$

```
n=100
z=rnorm(n)
y=rnorm(n,z,1)

data=list(y=y,z=z)
formula=y~1+z
result=inla(formula,family="gaussian",data=data,
       control.family=list(hyper = list(prec =
                           list(prior="loggamma",param=c(0.1,0.1)))))
```

Notes

None